

Generative Physics Informed Machine Learning Method for DC-Link Capacitance Estimation

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Abstract—This article proposes an innovative generative physics-informed machine learning (GPIML) method for the estimation of dc-link capacitance during the precharging process of the vehicular power systems, which contributes to greatly enhancing the reliability of electrified transportation. Different from the other machine learning-based estimation approaches, the proposed method produces highly accurate results using small input experimental dataset. To enable sufficient neural network training, diffusion algorithm is first adopted in the proposed method to augment the training data based on small input dataset. Then, the augmented data is fed to a physics-informed long short-term memory (PILSTM) algorithm to estimate the dc-link capacitance. Superior accuracy and strong robustness to measurement noises are achieved. The effectiveness of the proposed method is validated through experimental studies.

Index Terms—Capacitance estimation, dc-link capacitor, electrified transportation, generative machine learning, physic informed machine learning.

I. INTRODUCTION

DC-LINK capacitors are ubiquitous in modern electrified transportation systems, including the electric vehicles, traction drives, all electric ships and more electric aircrafts [1], [2], [3]. To obtain long lifespan, high reliability and low impedance, metalized film capacitor is increasingly preferred over the electrolytic capacitor in the aforementioned applications [4]. Despite its enhanced performance, the metalized film capacitor remains one of the most vulnerable components in the vehicular power systems [5]. To ensure the continuous and reliable operation of vehicular power systems, the capacitor

needs replacement when its capacitance degrades to a certain level. Therefore, the accurate estimation of capacitance is paramount.

In recent years, various parameter estimation methods have been proposed in the literature [6], [7], [8], [9], [10], [11], [12], [13]. For example, the current injection method is widely used to identify the capacitance of dc-link capacitor [6], [7], [8]. In [6], a 30 Hz ac current is injected into the dc-link capacitor. Even though accurate estimation results are reported, the method can not be utilized in the online testing (or load testing) and it inevitably increases the current harmonics. Besides, a Goertzel algorithm based current injection method is presented to estimate the capacitance of dc-link capacitor [7]. Compared with the fast Fourier transformation (FFT) and discrete Fourier transformation (DFT), the Goertzel algorithm is computationally more efficient with less memory usage. Nevertheless, an extra sensor is required to measure the capacitor current, which increases the hardware cost. In [8], a low-frequency current is injected into the back-to-back ac/dc and dc/ac PWM converter to estimate the dc-link capacitance without using extra current sensor. Its harmonics are effectively attenuated to meet the IEEE 519-2014 standard [8]. Nonetheless, the frequency and amplitude of the injected signal are selected by trial and error, which increases the design difficulty as a systematical design guideline is not provided. In addition to the current injection method, some energy dispatching or physics model based methods are presented in the literature. For instance, a power dispatching method is reported [9]. Specifically, the power dispatched in the capacitor is modeled as a function of the capacitance, the dc-link voltage, and its derivative. Unfortunately, this method is highly sensitive to noises because it directly employs the derivative term in the calculation. In [10], the estimated capacitance is formulated as a function of an electrical model involving two voltages sampled at the two adjacent zero-crossing points of average capacitor current and the net change in capacitor charge. In theory, this method is also sensitive to sampling noises. To mitigate the influence of noises, an extended Kalman filter (EKF) based estimation method is proposed to iteratively estimate the capacitance [11]. Although the EKF gives excellent performance, its model dependence and the requirement of covariance matrix tuning make it less attractive in practical applications. In [12], a recursive least square (RLS) based capacitance estimation algorithm is proposed. This method

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employs a low-pass filter to attenuate the impact of measurement noises. However, similar to the EKF, fine-tuning work is needed to find a suitable covariance matrix that ensures excellent performance of RLS [13]. Moreover, to have fast enough convergence, the initial weighting matrix of RLS also needs to be empirically tuned. An improved RLS called recursive extended least square (RELS) is reported [14]. Even though a faster response was reported, the weighting matrix tuning issue remains unsolved.

To eliminate the model dependence problem and the tedious observer gain tuning process, machine learning techniques are proposed for dc-link capacitance estimation [15], [16], [17]. For instance, a gray wolf optimization assisted back propagation (BP) algorithm is proposed for dc-link capacitance estimation in electric vehicles [18]. Notwithstanding, massive sets of the data are required to train neural network in this method, which can be costly and time-consuming. In practice, the operating environment and conditions of power converters are unpredictable. This greatly increases the difficulty of data acquisition for machine learning algorithms because the requirement of getting sufficiently large training datasets implies the necessity of carrying out experiments under various conditions. Unfortunately, without sufficient training data, the performance of machine learning algorithms can be apparently degraded. Another important issue of the training dataset is measurement noise, which can have comparatively high amplitudes in traction and locomotive systems [19], [20]. One of the feasible ways to ease the stringent demand on training data is to combine the physical model of the target system and the machine learning technique, which is also known as physics informed machine learning (PIML) [5], [21]. In [22], a PIML method is proposed to implicitly solve the characteristic equations of components in dc/dc buck converter. This method employs implicit Runge-Kutta algorithm to solve the latent states between two consecutive samples, so that the training dataset is augmented. Even though accurate estimation is achieved with the presence of sampling noises, this method experiences a heavy computational burden. Another way to expand the existing limited dataset is to utilize the generative model to produce more data traces that are qualified to be used as the training data. For example, the generative adversarial network (GAN) is proposed to extend the training dataset for neural network based battery aging monitoring [23]. Excellent training outcome is reported. However, due to the differences in sampling rate, data features, and design objectives, this method is not directly applicable to the estimation of dc-link capacitance in vehicular power systems.

To address the aforementioned problems in the dc-link capacitance estimation, this article proposes a generative physics informed machine learning (GPIML) technique for capacitance estimation. Specifically, the proposed method combines the generative learning and the PIML models, where the generative model is used to effectively extend the training dataset and PIML model is trained to learn the mapping relationship between the sampled dc-link voltage and capacitance. The contributions of this work can be summarized as follows.

- 1) It proposes a novel scheme to accurately estimate the dc-link capacitance using only small experimental dataset (only 50 data traces in this dataset).
- 2) A generic diffusion model is employed as the generative model, which provides excellent data augmentation performance with a low complexity Unet structure compared with the GAN. Moreover, the diffusion model is cross checked by the physical model of capacitor to ensure that the generated data is physically meaningful while matching the other probabilistic properties of the experimental data. The generated data was cross-checked against the capacitor partial differential equations outlined (4) before being fed into the PILSTM model.
- 3) A physics informed long short-term memory (PILSTM) model is trained based on the augmented dataset. Inherited from the long short-term memory (LSTM), the proposed PILSTM model learns not only the statistical properties but also the physical characteristics from the training dataset. It produces superior estimation results with noise-immunity capability.

The rest of this article is organized as follows. The model of dc-link capacitor in the traction systems is introduced in Section II. The proposed generative physics informed machine learning is presented in Section III. Then, Section IV demonstrates the detailed implementation of the proposed GPIML model and the configuration of the optimizer. After that, experimental results are illustrated in Section V to verify the efficacy of the proposed method. Finally, the conclusions are drawn in Section VI.

II. MODEL OF DC-LINK CAPACITOR IN TRACTION SYSTEMS

Fig. 1 shows the dc-link circuit of a typical inverter fed traction system. Physically, the relationship between the voltage and current of dc-link capacitor can be described as

$$C \frac{du(t)}{dt} = i(t) \quad (1)$$

where C is the capacitance, $u(t)$ is the voltage across the capacitor, and $i(t)$ is the current through the capacitor. Solving the above differential equation yields (2)

$$u(t) = E(1 - e^{-\frac{t}{RC}}) \quad (2)$$

where E is the precharging voltage and R is the lumped sum parasitic and precharging resistance. Notably, to ensure a smooth voltage ramp-up across the dc-link capacitor that leads to desirable data traces for the proposed machine learning algorithms, the precharge resistance should be significantly higher than the parasitic resistance of the dc-link capacitor. Then, the capacitance can be derived as

$$f(u, t, R) = C = \frac{-t}{R \ln \left(1 - \frac{u(t)}{E} \right)} \quad (3)$$

Taking the full-order partial differentiation of (3) yields the following full-order partial differential equation (PDE)

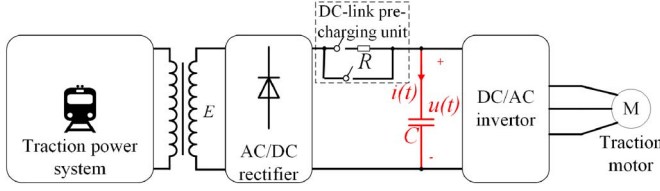


FIG. 1. Topology of the dc-link circuit in the traction converter system.

$$\begin{aligned}
 f_{\text{PDE}}(C, R, u, t) &= R \left(\frac{\partial C}{\partial u} \right)^2 \frac{\partial C}{\partial t} + CR \frac{\partial C^2}{\partial u \partial t} \frac{\partial C}{\partial u} - \left(\frac{\partial C}{\partial u} \right)^2 - CR \frac{\partial^2 C}{\partial u^2} \frac{\partial C}{\partial t} \\
 &= 0.
 \end{aligned} \quad (4)$$

From (4), it is difficult to directly estimate the capacitance due to the disturbances, sampling noises and electromagnetic interference.

III. GENERATIVE PHYSICS INFORMED MACHINE LEARNING

As discussed before, the integration of physical model of the target system into machine learning algorithms can help reduce the required training dataset. In this article, the dc-link capacitance can be calculated from the physical model (3). Note that R is the lumped sum parasitic resistance of capacitor, including both the equivalent series resistance and precharging resistance. u is the voltage across capacitor and E is the precharging dc-voltage, respectively.

Define a combination of data u , C and R as follows:

$$\Theta = \{[u(t), C, R] | f_{\text{PDE}}(C, R, u, t) \}. \quad (5)$$

Noticeably, a combination Θ_{abm} is said to be a solution of PDE (4) if and only if

$$\Theta_{\text{abm}} = \{[u(t), C, R] | f_{\text{PDE}}(C, R, u, t) = 0.\} \quad (6)$$

where $[u(t), C, R]$ is the analytical solution of PDE (4). Then, the set Θ_{abm} is a knowledge-based model. However, it is clear that the analytical solution is hard to obtain due to the presence of noises and the uncertainty of resistance in the real-time applications. Thus, this article proposes a GPIML model to estimate the dc-link capacitance, which can be conceptually illustrated using Fig. 2 below. In Fig. 2, the original sampled data is fed into the generative model, where the output is the augmented voltage data and the corresponding capacitance value. Then, the augmented dataset is used to train the physics informed machine learning.

A. Physics Informed LSTM

In this article, the LSTM model, which is a special recurrent neural network, is employed to estimate the dc-link capacitance. LSTM model adopts the gate-layer and cells to capture the input–output relationship statistically. Generally, its output can be mathematically expressed as

$$h_t = f(W(x_t, h_{t-1}) + b) \quad (7)$$

where x_t is the input layer. h_t and h_{t-1} are the hidden states of the slice at t th and $(t-1)$ th sample, which are also the outputs. $W = \{W_1, W_2\}$ is the network weighting matrix to be trained or fitted. b is the approximation bias. And ϕ is the activation function. In this article, the input is set to be $u(t)$ and the output is defined as the capacitance. Moreover

$$h_t = \mathbb{O}_t \cdot \phi_h(S_t) \quad (8)$$

where S_t is the cell state at time t

$$S_t = G_t S_{t-1} + \tilde{S}_t M_t \quad (9)$$

where $\tilde{S}_t$ denotes the latent cell state at time t . M_t and G_t are the memory and forget gate-layers at time t , which are defined as follows:

$$\begin{aligned}
 \tilde{S}_t &= \phi_h(W_c(x_t, h_{t-1}) + b_1) \\
 M_t &= \phi_s(W_m(x_t, h_{t-1}) + b_2) \\
 G_t &= \phi_s(W_g(x_t, h_{t-1}) + b_3) \\
 \mathbb{O}_t &= \phi_s(W_o(x_t, h_{t-1}) + b_4)
 \end{aligned} \quad (10)$$

where ϕ_h is the hyperbolic activation function. ϕ_s is the Sigmoid forgetting function. $b_{1,2,3,4}$ are the approximation bias, respectively. And the weighting vector $[W_c, W_m, W_g]$ contains the weightings of gate-layers to be trained. It can be seen that the i th output is recurrently updated from the current experience (i.e. memory gate layer) and the previous knowledge (i.e. forget gate-layer).

The combination of (7) to (10) yields the LSTM in (11)

$$\begin{aligned}
 \bar{C}_t &= f(W(x_t, h_{t-1}) + b) \\
 &= \phi_s \{W[x_t, \mathbb{O}_{t-1} \cdot \phi_h(S_{t-1})] + b\} \\
 &= \phi_s \left\{ W \left[x_t, \mathbb{O}_{t-1} \cdot \phi_h \left(G_{t-1} S_{t-2} + \tilde{S}_{t-1} M_{t-1} \right) \right] + b \right\} \\
 &= \phi_s \{W_1 x(u, t) + W_2 h_{t-1} + b\}
 \end{aligned} \quad (11)$$

where W_1 and W_2 are the network weightings to be trained.

Substituting (11) into (4) will yield the following (12):

$$\begin{aligned}
 f(\bar{C}, R, u, t) &= R \left(\frac{\partial \bar{C}}{\partial u} \right)^2 \frac{\partial \bar{C}}{\partial t} + \bar{C} R \frac{\partial \bar{C}^2}{\partial u \partial t} \frac{\partial \bar{C}}{\partial u} - \left(\frac{\partial \bar{C}}{\partial u} \right)^2 - \bar{C} R \frac{\partial^2 \bar{C}}{\partial u^2} \frac{\partial \bar{C}}{\partial t} \\
 &= c
 \end{aligned} \quad (12)$$

where c is a constant. Since the approximation bias is theoretically unavoidable, c is always a nonzero value. Now, define the network based model set as

$$\Theta_{\text{nbm}} = \{[u(t), \bar{C}, R] | f_{\text{PDE}}(\bar{C}, R, u, t) = c\} \quad (13)$$

where Θ_{nbm} is an observation of Θ . The resultant observation error is

$$L_r = \Theta_{\text{nbm}} - \Theta_{\text{abm}}. \quad (14)$$

Similarly, the dc-link capacitance estimation error is

$$L_C = \bar{C} - C. \quad (15)$$

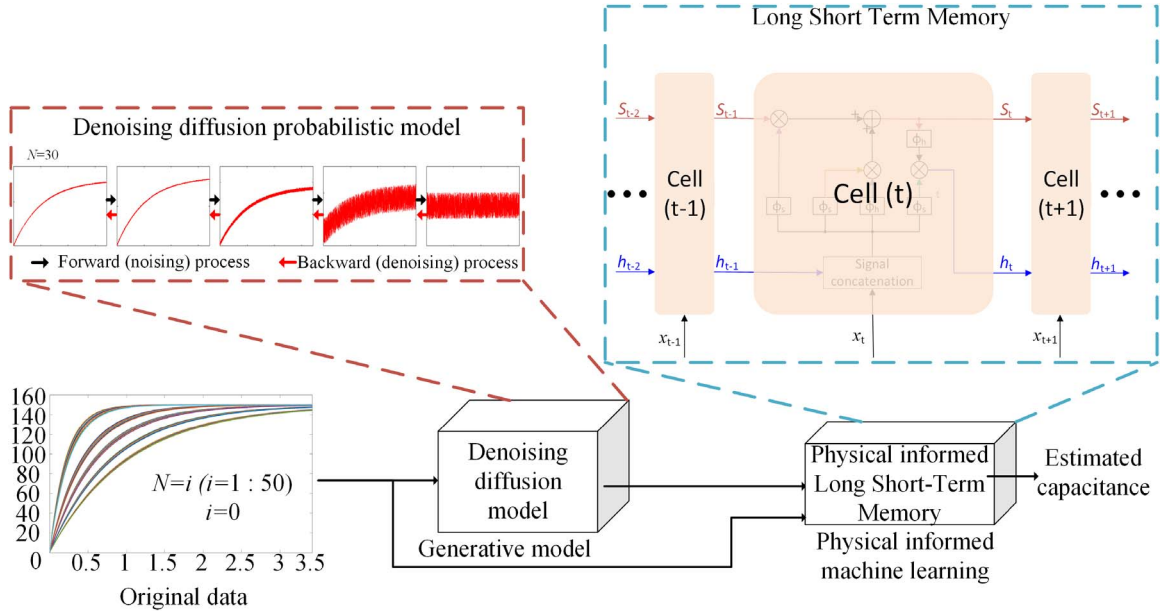


Fig. 2. Configuration of the proposed generative physics informed machine learning model.

The training process of PINN aims to find the network weightings that minimize the loss function defined in the mean square error (MSE) form as follows:

$$V(\bar{C}) = \text{MSE}(L_C) + \text{MSE}(L_r) \\ = \frac{1}{N} \sum |\bar{C} - C|^2 + \frac{1}{N} \sum |\Theta_{\text{nbm}} - \Theta_{\text{abm}}|^2. \quad (16)$$

And the training objective can be expressed as

$$W_{\text{PILSTM}} = \underset{W_1, W_2}{\text{argmin}} V(\bar{C}) \\ = \underset{W_1, W_2}{\text{argmin}} \left(\frac{1}{N} \sum |\bar{C} - C|^2 + \frac{1}{N} \sum |\Theta_{\text{nbm}} - \Theta_{\text{abm}}|^2 \right). \quad (17)$$

It is noteworthy that, in the loss function (16), the first term in the right-hand side of function represents the conventional loss function, which is supervised to fit the expected output. The second term of (16) is associated with physics informed learning because it introduces the PDE of capacitor circuit. Consequently, the training objective (17) guides the neural network to learn both the statistical relationship and the physical relationship between the input and output. However, the training process still requires a large amount of data (usually 100 000 data tracks or more) to optimize the network weightings. This requires extensive experimental work for data acquisition, which is quite costly and time-consuming.

B. Diffusion Model-Based Generative Learning

One cost-effective solution to the aforementioned high data demand problem is to utilize data augmentation technology to expand the existing dataset, instead of carrying out more experiments. Compared with traditional methods that estimate

the unknown data points within known data ranges, the denoising diffusion probabilistic model (DDPM) uses a statistical approach called variational Bayesian inference. This approach helps the model create new data points by learning to add and then remove noise from the data. It functions as a dynamically adjusted neural network known as an adaptive auto-encoder. In theory, the DDPM structure contains two parts, i.e. encoder (forward diffusion) and decoder (backward denoising). The encoder component can be formatted into a Markov-chain model as

$$x_{i+1} = \sqrt{a_{i+1}}x_i + \sqrt{1 - a_{i+1}}\mathbb{N}_i \quad (18)$$

where sequence $a = \{0 < a_1 < a_2 < a_3 < \dots < a_i < 1\}$ is the noise schedule. $\mathbb{N}_i = \mathcal{N}(0, 1)$ follows a normal Gaussian distribution.

Moreover, the distribution of data with added noise can be expressed as

$$p(x_{i+1}|x_i) = \mathcal{N}(\sqrt{a_{i+1}}x_i, (1 - a_{i+1})). \quad (19)$$

According to (18) and (19), the distribution function from x_0 to x_i can be written as

$$\begin{aligned} x_i &= \sqrt{a_i}x_{i-1} + \sqrt{1 - a_i}\mathbb{N}_{i-1} \\ &= \sqrt{a_i} \left\{ \sqrt{a_{i-1}}x_{i-2} + \sqrt{1 - a_{i-1}}\mathbb{N}_{i-2} \right\} + \sqrt{1 - a_i}\mathbb{N}_{i-1} \\ &= \sqrt{a_i} \left\{ \sqrt{a_{i-1}} \left[\sqrt{a_{i-2}}x_{i-3} + \sqrt{1 - a_{i-2}}\mathbb{N}_{i-3} \right] \right. \\ &\quad \left. + \sqrt{1 - a_{i-1}}\mathbb{N}_{i-2} \right\} + \sqrt{1 - a_i}\mathbb{N}_{i-1} \\ &= \sqrt{a_i a_{i-1} a_{i-2}} \times x_{i-3} + \sqrt{a_i a_{i-1}} \times (1 - a_{i-2})\mathbb{N}_{i-3} \\ &\quad + \sqrt{a_i} \times (1 - a_{i-1})\mathbb{N}_{i-2} + \sqrt{(1 - a_i)}\mathbb{N}_{i-1} \\ &= \prod_{k=0}^i \{\sqrt{a_k}x_0\} \end{aligned}$$

$$+ \mathbb{N}_i \cdot \sqrt{\sum_{n=0}^{i-1} \left\{ \prod_{k=1}^i \left[a_k (\sqrt{1-a_n})^2 \mathbb{N}_n \right] \right\} + \sqrt{(1-a_i)^2}} \quad (20)$$

It can be seen that, in (20), the second term of the right-hand side is the total sum of a series of Gaussian noise. Thus, from the law of large numbers, the sum of a series of Gaussian noise can be modeled as one generic Gaussian noise, which means that the (20) can be reformulated as

$$x_i = \sqrt{\hat{a}_i} x_0 + \sqrt{\beta_i} \mathbb{N} = \sqrt{\hat{a}_i} x_0 + \sqrt{1 - \hat{a}_i} \mathbb{N} \quad (21)$$

where $\hat{a}_i = \prod_{k=0}^i a_k$ and $\beta_i = 1 - \prod_{k=0}^i a_k$.

Hence, the following distribution holds:

$$p(x_i | x_0) = \mathcal{N}(\sqrt{\hat{a}_i} x_0, \beta_i) \quad (22)$$

where

$$\hat{a}_i^2 + \beta_i^2 = 1. \quad (23)$$

Up to now, it can be seen that the encoder aims to gradually add the Gaussian noise to the original data until it becomes a random noise. However, the purpose of DDPM is to generate the data from a random normal Gaussian noise. Thus, the decoder (i.e. denoising process) focuses on reconstructing the x_{i-1} from the posterior probability (24)

$$p(x_{i-1} | x_i, x_0) = \mathcal{N}(E(x_t, x_0), \text{VAR}) \quad (24)$$

where $E(x_t, x_0)$ and VAR are the expectation and variance of the posterior probability, respectively. They are trainable parameters. Thus, the posterior (24) can be reexpressed as (25) based on the Bayesian rule

$$p(x_{i-1} | x_i, x_0) = p(x_i | x_{i-1}, x_0) \frac{p(x_{i-1} | x_0)}{p(x_i | x_0)}. \quad (25)$$

Since the diffusion process is a Markov chain and the distribution function from x_0 to x_i follows (22), (25) can be modeled as

$$p(x_{i-1} | x_i, x_0) = \frac{\mathcal{N}\{\sqrt{a_i} x_{i-1}, (1-a_i)\} \mathcal{N}(\sqrt{\hat{a}_{i-1}} x_0, \beta_{i-1})}{\mathcal{N}(\sqrt{\hat{a}_i} x_0, \beta_i)}. \quad (26)$$

Combining (25), (26), and (22), the probability density function (PDF) can be written as

$$\begin{aligned} f(x_{i-1} | x_i, x_0) &= \frac{1}{\sqrt{2\pi(1-a_i)}} \exp \left\{ -\frac{(x_i - \sqrt{a_i} x_{i-1})^2}{2(1-a_i)} \right\} \\ &\cdot \frac{1}{\sqrt{2\pi\beta_{i-1}}} \exp \left\{ -\frac{(x_{i-1} - \sqrt{\hat{a}_{i-1}} x_0)^2}{2\beta_{i-1}} \right\} \\ &\div \left\{ \frac{1}{\sqrt{2\pi\beta_i}} \exp \left\{ -\frac{(x_i - \sqrt{\hat{a}_i} x_0)^2}{2\beta_i} \right\} \right\}. \end{aligned} \quad (27)$$

By the definition of PDF, (26) can be rewritten into the form of (28), as shown in the bottom of this page. Note that the constant gain $(1/\sqrt{2\pi})$ was ignored in the first step of (28), but this term is recalled in the last step to complete the distribution. The variation and mean of such a denoising process are described as (29) and (30), respectively

$$\text{Var} = \frac{(1-a_i)(1-\hat{a}_{i-1})}{1-\hat{a}_i} \quad (29)$$

$$E = \frac{(1-\hat{a}_{i-1})\sqrt{a_i} x_i + (1-a_i)\sqrt{\hat{a}_{i-1}} x_0}{1-\hat{a}_{i-1}}. \quad (30)$$

Note that the training process aims to find the variation and mean of (28). It is pointed out that the variation given in (29) is independent to the initial input x_0 . But mean expressed in (30) is related to the unknown initial input x_0 . Thus, the training process adopts the variational Bayesian inference to approximate

$$\begin{aligned} p(x_{i-1} | x_i, x_0) &\propto \exp \left\{ -\left[\frac{(x_i - \sqrt{a_i} x_{i-1})^2}{2(1-a_i)} + \frac{(x_{i-1} - \sqrt{\hat{a}_{i-1}} x_0)^2}{2(1-\hat{a}_{i-1})} - \frac{(x_i - \sqrt{\hat{a}_i} x_0)^2}{2(1-\hat{a}_i)} \right] \right\} \\ &= \exp \left\{ -\frac{1}{2} \left[-\frac{2\sqrt{a_i} x_i x_{i-1}}{1-a_i} + \frac{a_i x_{i-1}^2}{1-a_i} + \frac{x_{i-1}^2}{1-\hat{a}_{i-1}} - \frac{2\sqrt{\hat{a}_{i-1}} x_{i-1} x_0}{1-\hat{a}_{i-1}} + \frac{x_i^2}{1-a_i} + \frac{\hat{a}_{i-1} x_0^2}{1-\hat{a}_{i-1}} - \frac{(x_i - \sqrt{\hat{a}_i} x_0)^2}{1-\hat{a}_i} \right] \right\} \\ &= \exp \left\{ -\frac{1}{2} \times \left(\frac{1-\hat{a}_i}{(1-a_i)(1-\hat{a}_{i-1})} \right) \times \left[x_{i-1}^2 - 2 \left(\frac{\frac{\sqrt{a_i} x_i}{1-a_i} + \frac{\sqrt{\hat{a}_{i-1}} x_0}{1-\hat{a}_{i-1}}}{\frac{1-\hat{a}_i}{(1-a_i)(1-\hat{a}_{i-1})}} \right) x_{i-1} \right. \right. \\ &\quad \left. \left. + \frac{x_i^2(1-\hat{a}_{i-1})(1-\hat{a}_i) + \hat{a}_{i-1} x_0^2(1-a_i)(1-\hat{a}_i) - (x_i - \sqrt{\hat{a}_i} x_0)^2(1-a_i)(1-\hat{a}_i)}{(1-\hat{a}_i)^2} \right] \right\} \\ &= \exp \left\{ -\frac{1}{2} \times \left(\frac{1-\hat{a}_i}{(1-a_i)(1-\hat{a}_{i-1})} \right) \times \left[x_{i-1}^2 - 2 \left(\frac{(1-\hat{a}_{i-1})\sqrt{a_i} x_i + (1-a_i)\sqrt{\hat{a}_{i-1}} x_0}{1-\hat{a}_{i-1}} \right) x_{i-1} \right. \right. \\ &\quad \left. \left. + \left(\frac{(1-\hat{a}_{i-1})\sqrt{a_i} x_i + (1-a_i)\sqrt{\hat{a}_{i-1}} x_0}{(1-\hat{a}_i)} \right)^2 \right] \right\} \\ &\propto \mathcal{N} \left\{ \frac{(1-\hat{a}_{i-1})\sqrt{a_i} x_i + (1-a_i)\sqrt{\hat{a}_{i-1}} x_0}{1-\hat{a}_{i-1}}, \frac{(1-a_i)(1-\hat{a}_{i-1})}{1-\hat{a}_i} \right\}. \end{aligned} \quad (28)$$

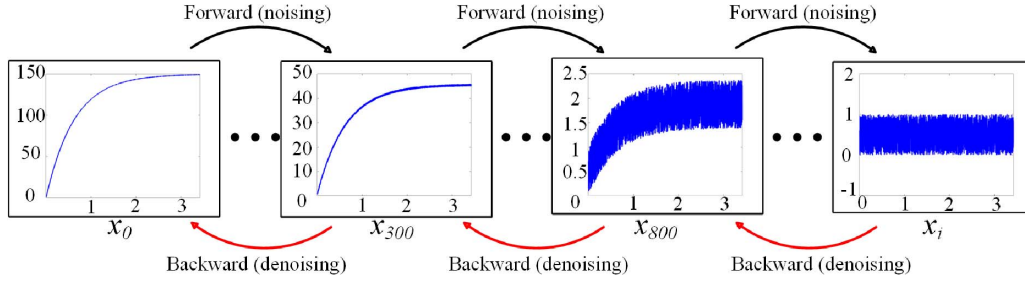


Fig. 3. Graphical illustration of the diffusion model.

the distribution (28). Moreover, from [24], the Kullback-Leibler (KL) divergence between the unknown normal distributions P , whose probabilistic distribution will be approximated by the trained U-net, and a given normal distribution Q can be defined as follows:

$$D_{KL}(P||Q) = \ln \left(\frac{\sigma_Q}{\sigma_P} \right) + \frac{\sigma_P^2 + (\mu_P - \mu_Q)^2}{2\sigma_Q^2} - \frac{1}{2} \quad (31)$$

where σ_P and σ_Q are the standard deviations of P and Q . μ_P and μ_Q are the means of P and Q . By rearranging the terms in (21), (32) can be derived

$$x_0 = \frac{x_i - \sqrt{1 - \hat{a}_i} \mathbb{N}}{\sqrt{\hat{a}_i}}. \quad (32)$$

Substituting (32) into the denoising mean E , yields

$$\begin{aligned} \mu_P &= \frac{(1 - \hat{a}_{i-1})\sqrt{a_i}x_i + (1 - a_i)\sqrt{\hat{a}_{i-1}}x_0}{1 - \hat{a}_{i-1}} \\ &= \frac{(1 - \hat{a}_i)x_i}{(1 - \hat{a}_i)\sqrt{a_i}} - \frac{(1 - a_i)\mathbb{N}}{\sqrt{1 - \hat{a}_i}\sqrt{a_i}} \\ &= \frac{x_i}{\sqrt{a_i}} - \frac{(1 - a_i)\mathbb{N}}{\sqrt{1 - \hat{a}_i}\sqrt{a_i}}. \end{aligned} \quad (33)$$

Then, the training objective of parameter neural network, i.e., $\bar{\mathbb{N}}$, can be expressed as

$$\begin{aligned} \argmin_{W_{DDPM}} D_{KL}(P||Q) &= \argmin_{W_{DDPM}} D_{KL}(\bar{\mathbb{N}}||\mathbb{N}) \\ &= \argmin_{W_{DDPM}} \left\{ \left\| \frac{1}{\sqrt{2\mu^2}} \left(\frac{(1 - \hat{a}_{i-1})\sqrt{a_i}x_i + (1 - a_i)\sqrt{\hat{a}_{i-1}}\bar{x}_0}{1 - \hat{a}_{i-1}} \right. \right. \right. \\ &\quad \left. \left. - \frac{(1 - \hat{a}_{i-1})\sqrt{a_i}x_i + (1 - a_i)\sqrt{\hat{a}_{i-1}}x_0}{1 - \hat{a}_{i-1}} \right\|^2 \right\} \\ &= \argmin_{W_{DDPM}} \left\{ \frac{1}{2\mu^2} \times \frac{(1 - a_i)^2 \hat{a}_{i-1}}{(1 - \hat{a}_{i-1})^2} \times \|\bar{x}_0 - x_0\|^2 \right\}. \end{aligned} \quad (34)$$

Substituting (32) into (34) yields the final training objective function of the denoising process

$$\begin{aligned} W_{DDPM} &= \argmin_{W_{DDPM}} \left\{ \frac{1}{2\mu^2} \times \frac{(1 - a_i)^2 \hat{a}_{i-1}}{(1 - \hat{a}_{i-1})^2} \times \|\bar{x}_0 - x_0\|^2 \right\} \\ &= \argmin_{W_{DDPM}} \left\{ \frac{1}{2\mu^2} \times \frac{(1 - a_i)^2 \hat{a}_{i-1}}{(1 - \hat{a}_{i-1})^2} \right. \\ &\quad \left. \times \left\| \frac{x_i}{\sqrt{a_i}} - \frac{(1 - a_i)\bar{\mathbb{N}}}{\sqrt{1 - \hat{a}_i}\sqrt{a_i}} - \frac{x_i}{\sqrt{a_i}} + \frac{(1 - a_i)\mathbb{N}}{\sqrt{1 - \hat{a}_i}\sqrt{a_i}} \right\|^2 \right\} \end{aligned}$$

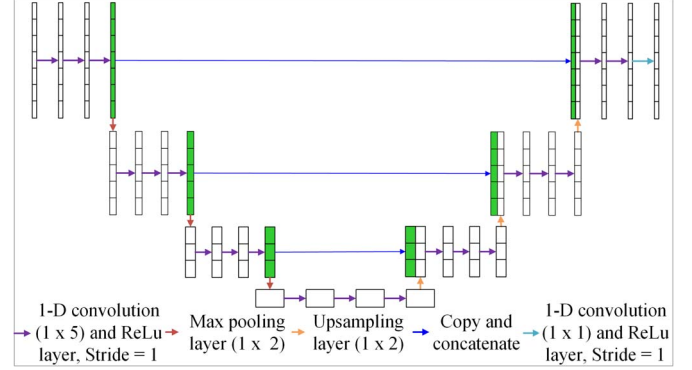


Fig. 4. Configuration of the U-net used in backward process of diffusion model.

$$= \argmin_{W_{DDPM}} \left\{ \frac{1}{2\mu^2} \times \frac{(1 - a_i)^2}{(1 - \hat{a}_i)a_i} \|\mathbb{N} - \bar{\mathbb{N}}\|^2 \right\}. \quad (35)$$

From (35), it can be seen that the parameter neural network (i.e., U-net) is trained to approximate the noise distribution from a given $\mathbb{N}$. This implies the combination of the variational Bayesian inference and a typical U-net is good enough for this application. In the forward noising procedure, the noise scheduler mixes Gaussian random noise into the sampled data, as shown (20). With this series of noised data, the backward denoising process adopts variational Bayesian inference, which provides a robust framework for estimating the underlying probabilistic distribution of the data. Specifically, advanced neural network architectures, such as U-Net with skip connections, help preserve detailed information during denoising stages, ensuring accurate capture of both global and local data features.

IV. NETWORK CONFIGURATION AND IMPLEMENTATION

For a better illustration of the proposed method, the configurations of the diffusion and LSTM networks as well as their training procedures are detailed in this section. First, in the data augmentation process, diffusion model is employed, which consists of forward noising and backward denoising steps. On the one hand, the configuration of forward noising diffusion model is depicted in Fig. 3. It is visible that Gaussian noises are purposely added to the initial data samples in the forward process according to (20). A conventional U-net is adopted to implement the variational Bayesian inference in the backward process due to its extraordinary performance of single-dimension time-series data process. Specifically, the U-net as

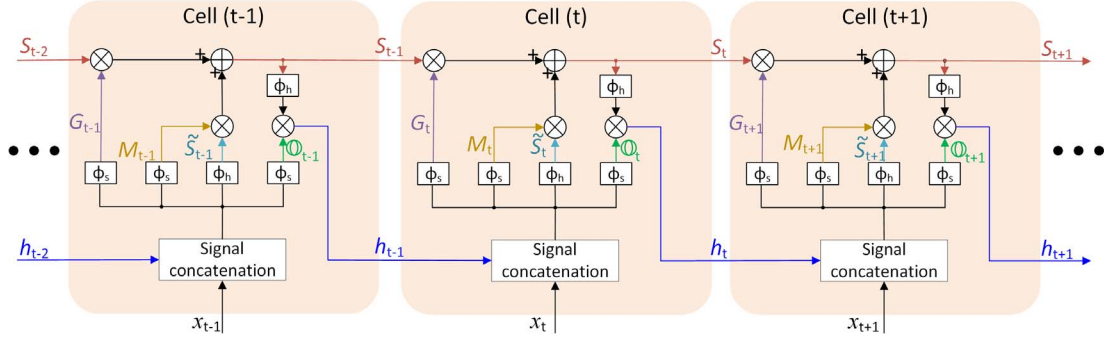


Fig. 5. Configuration of the PILSTM model.

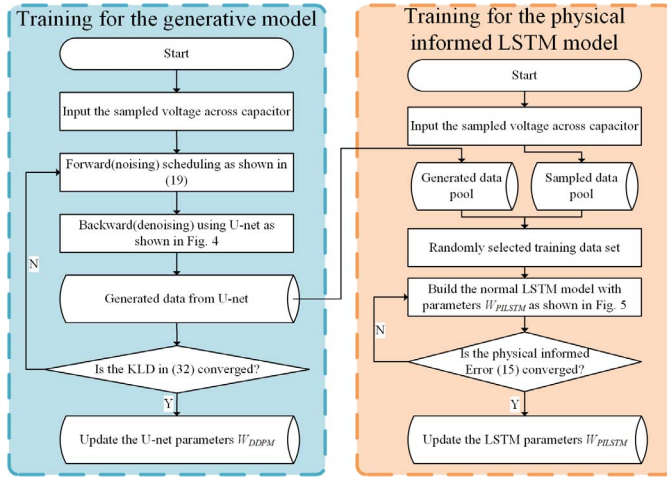


Fig. 6. Flowchart of offline training process.

shown in Fig. 4 is trained to approximate the posterior distribution from a given N . After data augmentation, the extended dataset is fed into the LSTM network to produce the capacitance estimation. The basic configuration of LSTM is shown in Fig. 5, where cells are cascaded and forget gate layers as well as the hidden states are used to connect cells.

From the previous section, the differential equations are introduced during the training process. Specifically, the physics informed error function (16) is used as the loss function to train the proposed PILSTM. The offline training process can be summarized in Fig. 6. On the left, the generative model training begins with inputting sampled voltage data. It proceeds with forward noising, and subsequently backward denoising using a U-net structure. The convergence of KL between the original training dataset and the Gaussian noises is continuously checked to update the U-net parameters. On the right, the PILSTM model training involves inputting voltage data, utilizing both generated and sampled data pools, selecting training datasets randomly, building the LSTM with parameters W , checking for convergence of the physics-informed error, and updating the LSTM parameters.

V. EXPERIMENTAL VALIDATION AND RESULTS

To validate the effectiveness of the proposed method, the hardware experiments were conducted on an experimental setup

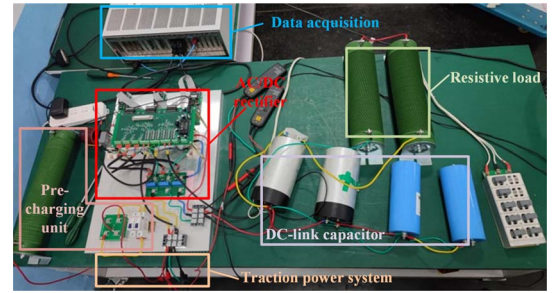


Fig. 7. Experimental setup of the dc-link capacitor in traction system.

TABLE I
PARAMETERS OF THE PROPOSED METHOD AND
TRAINING OPTIMIZER

Parameters	Value
Sampling frequency of the voltage	1kHz
LCR meter measuring frequency	100Hz
Ambient temperature	25° C
Traction power system voltage (RMS)	150 V
Precharging unit equal resistance	50 Ω
Initial learning rate of the optimizer	0.001
Number of hidden cells in PILSTM	50
Number of U-net layers	31
Number of DDPM noising sequence	1000

TABLE II
CAPACITANCE VALUE OF THE ORIGINAL SAMPLED DATASET

Capacitance Value (mF)	Variant Interval (%)	Group Range (mF)
1.203	8.3%	[1.203 ~1.303]
1.625	6.2%	[1.625 ~1.725]
2.410	4.1%	[2.410 ~2.510]
3.410	2.9%	[3.410 ~3.510]
4.402	2.3%	[4.402 ~4.502]

consisting of traction converter and metalized film capacitor, which is depicted in Fig. 7. Parameters of the experimental setup and the employed neural networks are listed in Table I.

The data acquisition process entailed the sampling and collection of original data utilizing the Dspace DS1007 data acquisition module. The proposed method is then trained on the sampled data using a regular laptop computer (AMD Ryzen

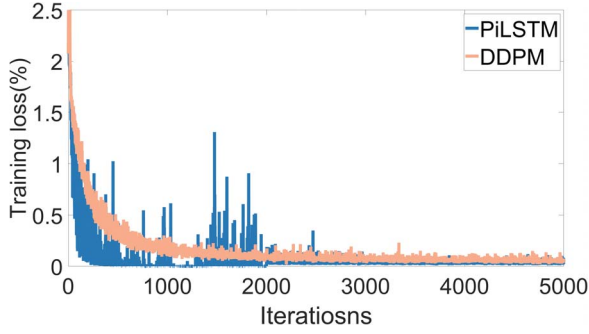


Fig. 8. Training loss of both models in the proposed method.

7 PRO 4750U, 1.7 GHz, 16 GB RAM). The value of the dc-link capacitance is delineated in Table II. The initially sampled dataset comprises a total of 50 distinct capacitance values, with the voltage across the capacitance being recorded over a duration of 3.42 s at a sampling frequency of 1 kHz. Consequently, the size of the complete original dataset is 50×3420 .

The stochastic gradient descent optimizer from [25] is adopted to train the neural networks as shown in Fig. 6 and the cosine annealing learning rate scheduler is used because it can lead to more effective learning and better model performance [26]. The training losses of the DDPM and PiLSTM are shown in Fig. 8. Noticeably, for the PiLSTM model, the training loss is defined as the objective function in (17), which quantifies the difference between the model's predictions and the actual values in the training dataset from both the statistical and physical perspectives. Similarly, for the DDPM model, the training loss is specified by the objective function in (35). It can be seen from Fig. 8 that both DDPM and PiLSTM converged after 2000 iterations.

A. Analysis of the DDPM

The trained DDPM is employed to augment the original dataset. As delineated in (28), the DDPM receives inputs conforming to a normal distribution, producing outputs that encompass augmented values of capacitor voltage and the corresponding capacitance. Fig. 9 illustrates a subset of 15 randomly selected trajectories from the augmented dataset. Upon comparative analysis with the original data as shown in Fig. 9, it is observed that the augmented dataset preserves the statistical and physical characteristics inherent to the original sampled data. This fidelity is attributed to the intrinsic learning capabilities of the DDPM. It employs a progressive learning process where noise is gradually added and then systematically removed, allowing the model to learn the data distribution in a stable and controlled manner. The use of variational Bayesian inference provides a robust framework for estimating the underlying data probabilistic distribution. The noise scheduling strategy and the specific training objective are designed to optimize the denoising and data reconstruction, enhancing the model's ability to learn and replicate data with high fidelity.

These intrinsic capabilities collectively contribute to the DDPM's superior performance in modeling and reproducing complex data distributions. As mentioned, the DDPM is used to

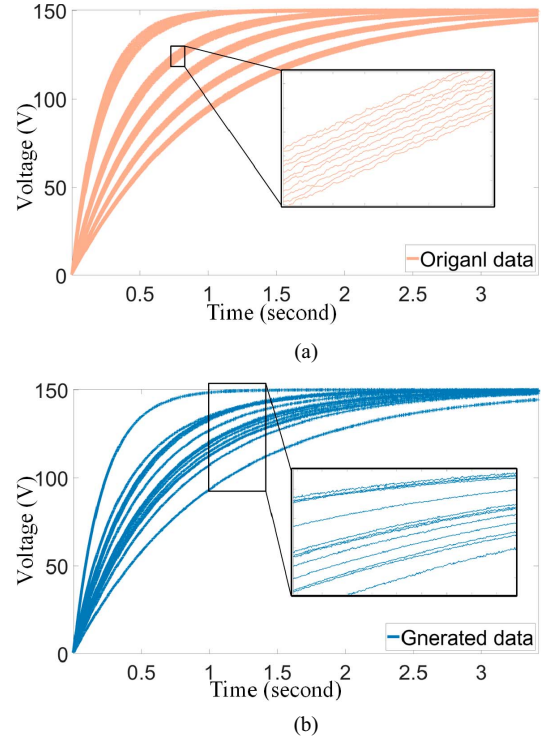


Fig. 9. Comparison of the original dataset and generated dataset. (a) Original dataset. (b) Output of DDPM.

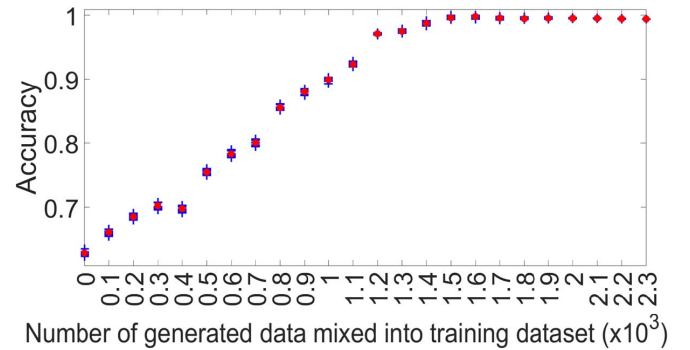


Fig. 10. Scatter plot of accuracy under different volume of the expanded training dataset: the mean of tests (red circle), accuracy of ten tests (blue cross).

expand the training dataset so that the GPIML can be properly trained. To verify the effectiveness, the box plot, i.e. Fig. 10, analysis illustrates that incorporating a larger volume of data generated by the DDPM into the training dataset significantly enhances the accuracy of GPIML model. Specifically, the ten testing scenarios were conducted under the different volume of expanded dataset and the median accuracy of each test is highlighted in red color. It can be seen that the median accuracy is 62.82% and maximum accuracy is only 63.49% if no generated data was mixed into the training dataset. With more generated data injected into the training set, the median is increased to more than 95%. After approximately 1500 points, the medians

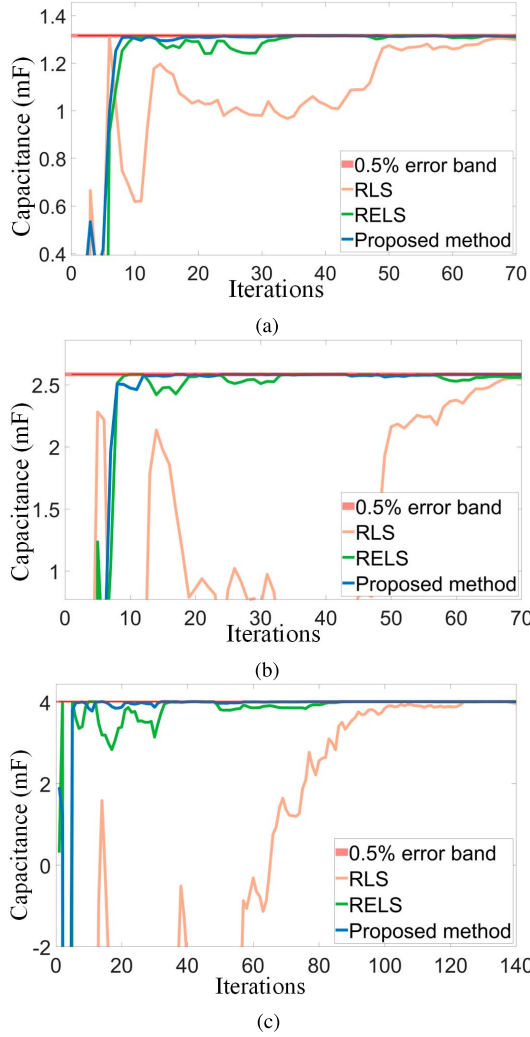


Fig. 11. Estimated capacitance and relative error under different methods: RLS method (orange color), RELS method (green color) and the proposed method (blue color). (a) Estimated capacitance of 1.319 mF. (b) Estimated capacitance of 2.583 mF. (c) Estimated capacitance of 4.014 mF.

reach a range that is more than 99.9% and the minimum accuracy is 99.8%. It is clear that, as more generated data is added, the median accuracy of the GPIML model increases and the variation of accuracy decreases, indicating improved reliability and consistency in model performance. However, beyond a certain threshold of generated data, it suggests a saturation point where additional data does not yield further improvements. The DDPM's ability to generate synthetic data enables the GPIML model to train on a broader range of datasets, enhancing its robustness and generalization capability.

B. Testing and Validation of GPIML

Subsequent to the completion of the offline training regimen, the GPIML architecture is employed for the estimation of capacitance based on the real-time capacitor voltage measurement. The GPIML has undergone rigorous training utilizing an augmented dataset, which is a synthesis of the original dataset

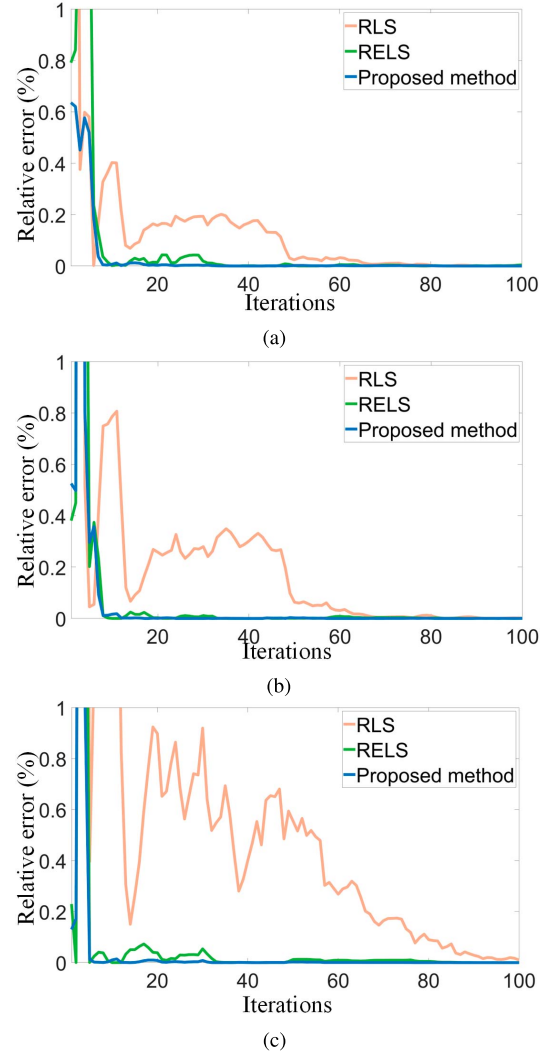


Fig. 12. Estimation relative error under different methods: RLS method (orange color), RELS method (green color) and the proposed method (blue color). (a) Relative error of 1.319 mF. (b) Relative error of 2.583 mF. (c) Relative error of 4.014 mF.

amalgamated with the outputs derived from the DDPM model. To show the high accuracy of the proposed GPIML, the capacitance estimation performances of the traditional recursive least squares (RLS) and recursive extended least squares (RELS) are compared with the proposed method. The results are illustrated in Figs. 11 and 12. Fig. 11 shows that the estimation error of the proposed method quickly converges to be within the acceptable error margin of 0.5% in approximately 20 iterations, whereas it takes the RELS and RLS algorithms over 30 and 70 iterations, respectively, to achieve this. Furthermore, the relative estimation error corresponding to the capacitance value of 1.319mF is depicted in Fig. 12. It is clear that the relative estimation error of the proposed method converges to the acceptable level faster compared with those of RELS and RLS. Similar to the scenario of estimating 1.319 mF capacitance, the proposed GPIML method demonstrates superior capacitance estimation performance over the RELS and RLS under the testing cases of 2.583 and 4.014mF. It is noteworthy that the capacitance values

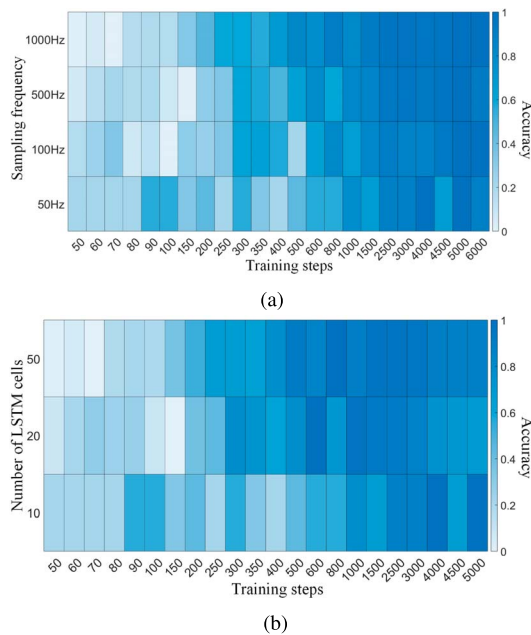


Fig. 13. Heatmaps of the accuracy under different hyperparameters. (a) Accuracy under different sampling frequencies. (b) Accuracy under different number of PILSTM cells.

in all three tests are not included in the sampled dataset as shown in Table II. Especially, the capacitance value 4.014mF is greater than the maximum capacitance in training data group 4 (i.e. 4th row in Table II) and less than the minimum value of training data group 5 (i.e. 5th row in Table II). The proposed method shows the superior performance with the unseen data, which indicates its generalization ability. This enhanced performance predominantly arises from the ability of the proposed GPIML framework on data generation in regions that are uncovered by the original dataset.

C. Hyperparameter Sensitivity Analysis and Robustness

The hyperparameters within the GPIML framework significantly influence the precision of estimation. For instance, reducing the sampling frequency of voltage measurements may degrade the accuracy. Furthermore, the quantity of PILSTM cells exerts a notable effect on the accuracy of the estimation. Therefore, an analysis of hyperparameter sensitivity is imperative and is presented in this section. The resultant accuracy heatmaps are depicted in Fig 13. It is important to note that the original dataset is down-sampled at various ratios to emulate the decrease of sampling frequency. First, the estimation accuracy under various sampling frequencies is depicted in Fig. 13. The results demonstrate that, subsequent to a sufficient volume of training iterations, the accuracy of the proposed method consistently exceeds 99.9%. Regarding the appropriate number of PILSTM cells, our observations show that accurate and robust capacitance estimation can always be obtained when the number cells exceeds 20. In theory, inadequate number of cells in the LSTM can cause underfitting and degraded performance [27].

VI. CONCLUSION

This article proposes a novel GPIML method for accurately estimating the dc-link capacitance, which contributes to fundamentally improve the reliability of electrified transportation systems. The GPIML approach, combining a generative model for data augmentation with a PILSTM model, demonstrates superior accuracy and robustness compared with the traditional methods, despite utilizing a small experimental dataset. The proposed method demonstrates to the power electronic industry that generative machine learning techniques can effectively extend the experimental dataset to reduce the cost and time on carrying out too many experiments. The efficacy of the proposed method is verified by experimental results and its comparison with the traditional capacitance estimation approaches.

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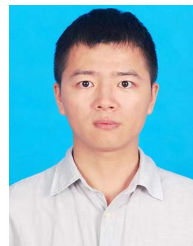
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